

Terms Link Analysis

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April 2026

Introduction

The present work realizes an analysis from a dataset composed of 153 documents, where each document presents some or the totality of 21 terms. The main objective is to explore and describe the data and plot a graph that shows the links between the terms. Text analysis is a widely used technique across many domains, enabling the extraction of meaningful patterns and hidden relationships from large collections of documents. In this context, link analysis and data visualization serve as key tools to understand how terms relate to one another and how they are distributed across documents.

To achieve this, we used Python along with several complementary libraries: pandas for data loading and manipulation, matplotlib and seaborn for statistical visualization, and NetworkX for constructing and visualizing the term connection graph. The analysis is structured in two main parts. First, we examine term frequencies, document distributions, and raise specific questions about the data; second, we perform a link analysis where term correlations are computed and represented as a graph, with node sizes reflecting how frequently each term appears across the data set.

Data Set Description

The data set consists of 153 documents, each represented as a row indexed by a document number. The 21 columns correspond to the following terms: advice, author, cold, computing, encourage, examples, information, help, lesson, news, opinion, positions, proposal, recommend, research, sentiment, slides, suggestion, tip, tutorial, and users. All values are binary (0 or 1), indicating the absence or presence of a given term in a document.

Exploratory Data Analysis

Term frequency distribution

The most common terms in the dataset are recommend, encourage, help, advice, and news, with, respectively, 69, 52, 46, 23 and 20 occurrences.

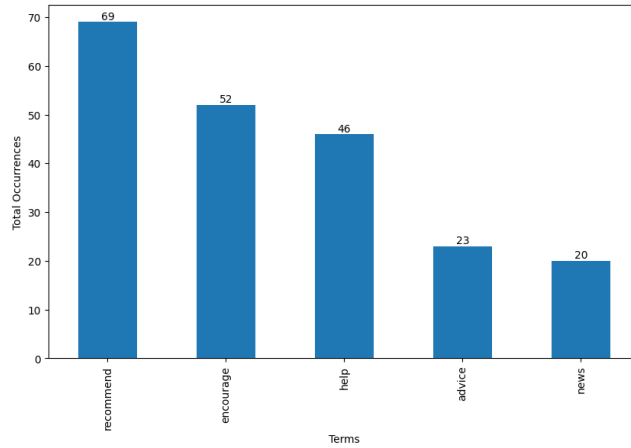
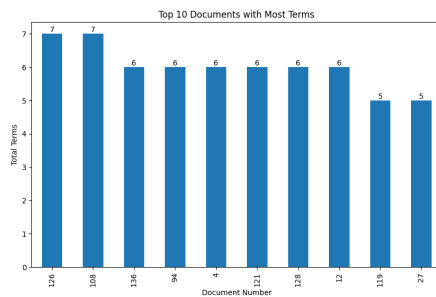


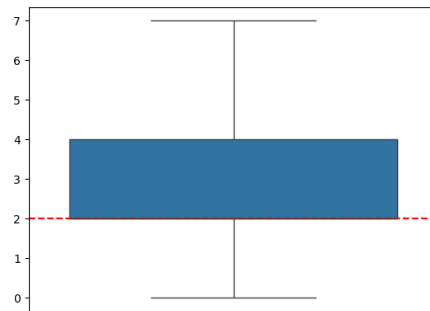
Figure 1: Most common terms

Document-level analysis

We examine how the terms are distributed across the 153 documents in the data set. The bar chart of the top 10 documents by total term count reveals that documents 108 and 126 are the richest in content, containing 7 terms. The remaining documents in the top 10 contain between 5 and 6 terms, and only 8 documents in total exceed the threshold of 5 terms, confirming that highly term-dense documents are rare.



(a) Documents with the highest number of terms



(b) Number of terms by document

The boxplot presents the median value as 2 and this indicates that 50% of the documents has 2 or less terms, and the other 50% has more then 2 terms by document. The main range of documents are between the first and third quartile, that means 2 and 4 terms by documents.

Key findings

The exploratory analysis allowed us to answer several concrete questions about the dataset. First, regarding term frequency, recommend is the most common term overall, appearing across 69 documents, followed by encourage (52) and help (46). These three terms alone account for a significant portion of all term occurrences.

At the document level, document 108 contains the highest number of terms with 7, making it the most term-rich entry in the dataset. On average, documents contain 2.6 terms, with a standard deviation of 1.5. Only 8 documents contain more than 5 terms, confirming that dense documents are the exception rather than the rule.

Term Correlation & Link Analysis

Correlation heatmap

To identify relationships between terms, a Pearson correlation matrix was computed across all 21 terms and visualized as a heatmap. The resulting figure reveals several notable patterns in how terms co-occur across documents.

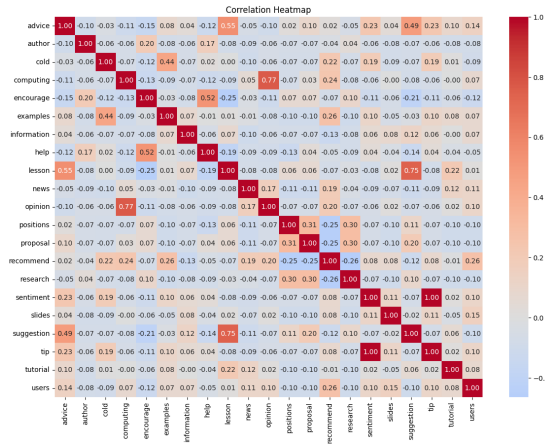


Figure 3: Correlation heatmap of terms

The strongest positive correlations are found between computing and opinion (0.77), lesson and suggestion (0.75), and advice and lesson (0.55). These pairs tend to appear together consistently, suggesting thematic groupings within the collection. For instance, computing and opinion may reflect documents discussing perspectives on technical topics, while lesson, suggestion, and advice cluster around instructional or guidance-oriented content.

On the other end, several term pairs show weak negative correlations, most notably positions and recommend (-0.25), proposal and recommend (-0.25), and encourage and suggestion (-0.21). While these values are modest in magnitude, they suggest that documents containing recommend tend not to use positions or proposal, pointing to distinct document types within the dataset.

Overall, the heatmap confirms that the term space is not uniformly connected — a small number of term pairs are strongly linked, while the majority share weak or near-zero correlations, consistent with the sparse nature of the dataset observed earlier.

Graph visualization & interpretation

To move beyond pairwise correlations and visualize the overall structure of term relationships, a graph was constructed using the NetworkX library. In this graph, each of the 21 terms is represented as a node, and an edge is drawn between two nodes whenever their Pearson correlation is non-zero. Meaning any detectable linear relationship between two terms, results in a connection. Node sizes were scaled proportionally to each term’s total frequency across the data set, computed as the sum of occurrences over all 153 documents. This means that high-frequency terms such as recommend, encourage, and help are rendered as larger nodes, providing an immediate visual cue for which terms dominate the dataset. Conversely, less frequent terms such as author, tip, and sentiment appear as smaller nodes.

In terms of visual prominence, recommend, encourage, and help appear as the largest nodes in the graph, reflecting their status as the most frequent terms in the dataset with 69, 52, and 46 occurrences respectively. At the other end, author, tip, sentiment, and opinion are rendered as the smallest nodes, consistent with their low frequency of only 8 occurrences each.

In the full graph, every node is connected to all 20 others, resulting in 210 edges. While this confirms that no term is entirely independent from the rest, the graph is too dense to draw meaningful structural conclusions — all nodes appear equally central, and the visualization offers limited interpretability beyond confirming the dataset’s overall connectivity.

Applying a threshold of 0.3 reduces the graph to 8 edges, immediately revealing a clearer structure. Three terms emerge as the most connected nodes: advice, lesson, and suggestion, each with 2 connections, forming a thematic clus-

ter around instructional and guidance-related content. The remaining edges link computing–opinion, encourage–help, cold–examples, positions–proposal, and sentiment–tip as isolated pairs. Notably, 7 terms become completely disconnected at this threshold — author, information, news, recommend, research, slides, tutorial, and users — indicating that despite being present in the data set, they do not strongly co-occur with any other term.

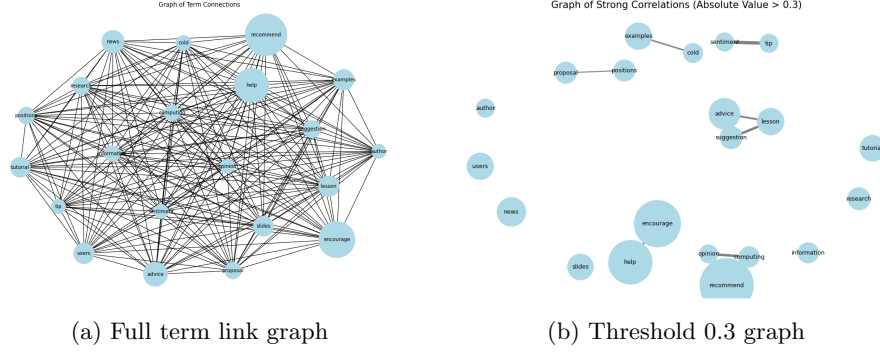


Figure 4: Comparison of term link graphs: showing the impact of applying a correlation threshold to clarify thematic structures.

One observation is that the most frequent term, *recommend*, became an isolated node at higher correlation thresholds. This suggests that while it is a "global" term used frequently across the dataset, it does not have a specific "local" partner it consistently appears with, unlike more specialized terms.

Conclusion

The analysis conducted in this report provides an overview of the term relationships within the provided dataset. Through data analysis, we identified a high degree of sparsity, with most documents containing only a few terms. A small subset of terms, specifically *recommend*, *encourage*, and *help*, dominate the collection, while most other terms appear sporadically. The link analysis transitioned from simple frequency counts to identifying thematic structures. By calculating Pearson correlations, we discovered meaningful pairings. A significant finding emerged when applying a correlation threshold of 0.3 to our network graph. This revealed a central cluster—*advice*, *lesson*, and *suggestion*—and several distinct thematic dyads such as *computing–opinion* and *positions–proposal*.

Thus with link analysis we can extract the information about how similar terms appears together.